

Information and Expression of Simultaneous and Backward Associations: Implications for Contiguity Theory

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Five conditioned lick-suppression experiments with water-deprived rats examined the possibility that simultaneous and backward associations are learned, but are not expressed as anticipatory responses in common indexes of associative strength. Experiments 1-4 used a sensory preconditioning procedure in which clicks preceded the onset of a tone. Subsequently, the tone was paired with footshock in either a forward, simultaneous, or backward arrangement. In no case did the tone trained in the simultaneous or backward manner elicit a conditioned response. However, Experiments 1, 2, and 3 determined that the clicks, which predicted the tone, evoked equally strong conditioned responses regardless of whether the tone was paired with the shock in a forward, simultaneous, or backward manner. Experiment 4 found that responding to the clicks was degraded following post-conditioning extinction of the tone, regardless of whether the tone had been paired with the shock in a forward or simultaneous manner. Experiment 5 determined that if the click and tone were paired simultaneously, the click failed a test for excitation following tone-shock simultaneous pairings but passed a test for excitation following tone-shock forward pairings. Collectively, these findings suggest that predictive information (i.e., a forward relationship between stimuli) is not necessary for the acquisition of an association, but may promote the expression of the association in an anticipatory response system. Moreover, these results suggest that associations are not simple linkages, but contain information regarding the temporal relationship of the associates. © 1988 Academic Press, Inc.

The apparent inability of simultaneous and backward conditioning preparations to support appreciable learning has been a persistent problem

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for contiguity-based theories of learning. In the initial systematic exploration of associative learning, Pavlov (1927, p. 27) concluded that a simultaneous or backward relationship between a conditioned stimulus (CS) and an unconditioned stimulus (US) was not sufficient to establish an excitatory association. This conclusion was problematic for Pavlov's view of conditioning, as Pavlov, like the British empiricist philosophers before him, believed that the repeated occurrence of two events in close temporal contiguity was both necessary and sufficient for the formation of an association.

A popular notion in recent years has been that the mere contiguous occurrence of two events is not sufficient for the formation of an association between their representations in memory. Rather, the CS must provide the subject with information concerning an impending reinforcer (e.g., Egger & Miller, 1962; Kamin, 1969; Rescorla, 1972, 1988; Rescorla & Wagner, 1972). This informational hypothesis has been applied most often to cases of competition between stimuli for associative strength, but it is equally applicable to the problems of simultaneous and backward conditioning (Cantor, 1981). As Mackintosh (1983, p. 209) has suggested, conditioning, from a functional standpoint, should provide the subject with information about the cause of a reinforcer. As simultaneous and backward relationships do not serve this function, Mackintosh concluded that their failure to support conditioning is not surprising.

Most informational accounts of learning maintain that mere contiguity between a CS and a US is not sufficient for the establishment of an association. However, this view encounters a number of problems when applied to the cases of simultaneous and backward conditioning. First, numerous results in the human and animal learning literature provide evidence that a forward relationship is not necessary for the establishment of an association between stimuli. For instance, human learning theorists place little emphasis on the temporal relationship between stimuli because strong associations are apparently formed between stimuli presented simultaneously in preparations such as the paired-associates task (e.g., Gibson, 1940). Within the animal literature, Speers, Gillan, and Rescorla (1980), as well as Rescorla (1980a), have reported that the association between simultaneously presented neutral stimuli is as strong or stronger than the association formed when those stimuli are presented sequentially. While Rescorla (1981) has suggested that these latter demonstrations of simultaneous associations may represent perceptual configuring as opposed to associative learning, the evidence for such a process is tenuous. Moreover, using a conditioned suppression procedure, Burkhardt and Ayres (1978) and Mahoney and Ayres (1976) have reported that, at least early in training, a CS paired simultaneously with a shock US will support substantial excitatory responding. Similar results have also been reported

in backward conditioning paradigms (e.g., Ayres, Haddad, & Albert, 1987).

Most damaging to information theory is recent evidence by Farley and Alkon (1987) and Grover and Farley (1987) demonstrating stronger simultaneous than forward conditioning in *Hermisenda*. In these studies, a light CS was paired simultaneously with high-speed rotation of the subject, and depolarization of the *Hermisenda's* photoreceptor to the light served as the index of associative strength. These results are particularly illuminating, given the choice of CS and response system. Rather than monitor a response that depends on the anticipation of a US for its expression and is several times removed from its neural substrates (e.g., conditioned suppression), these studies indexed associative strength directly, that is, prior to its translation into higher order response systems. Similar results have been reported in other studies where learning has been monitored at the neural level (e.g., Hawkins & Kandel, 1984; Kandel & Schwartz, 1982).

One alternative to conventional informational accounts of learning is the possibility that an association is formed between a simultaneously presented CS and US, but the CS does not evoke an *anticipatory* response (e.g., eyeblink, salivation, or suppression of ongoing behavior) because it does not bear a forward relationship to the US. The use of the term "anticipatory response" is not intended to imply that the CR necessarily serves any preparatory role (Perkins, 1968). Rather, the term anticipatory merely implies that the CS bears a forward temporal relationship with the US, and the conditioned response is in some sense representative of that predictive relationship. The possibility that the simultaneous and backward conditioning deficits represent an absence of an anticipatory response, as opposed to a learning failure per se, was similarly suggested by Holland (1984), who argued that these response failures are likely artifacts of our choice of response system, as opposed to an associative failure on the part of the animal.

EXPERIMENT 1

Experiment 1 was intended to determine whether a CS paired simultaneously with a US can acquire associative strength that is not expressed in a conventional excitatory test which is presumably sensitive to anticipatory responses (Cantor, 1981). In this study, a click and tone were first either forward-paired, that is, presented sequentially (Groups F-F and F-S), or unpaired (Groups UP-F, UP-S, and UP-UP) in the sensory preconditioning phase. Subsequently, two groups (F-F and UP-F) received tone-shock forward pairings and two groups (Groups F-S and UP-S) received the tone and shock paired simultaneously. The final group (UP-UP) received the tone and shock unpaired in this phase of the study. All groups were then tested for suppression of ongoing drinking, first in

the presence of the click and then in the presence of the tone. Preliminary work had determined that with the parameters employed here the tone would evoke little or no conditioned responding in groups that had received simultaneous tone and shock presentations (Groups F-S and UP-S) and would evoke a strong conditioned response in groups that received forward pairings (Groups F-F and UP-F). Our central interest was in the response to the clicks when the clicks predicted the onset of the tone (Groups F-F and F-S). If the tone was associated with the shock as a result of the simultaneous tone and shock presentations and if this association was not expressed as an anticipatory response because the tone lacked a forward temporal relationship to the shock, then the click, which was trained with a forward relationship to the tone, might be expected to evoke an excitatory conditioned response.

Method

Subjects. Forty-eight male and 48 female, naive, Sprague-Dawley-descended, rats obtained from our own breeding colony served as subjects. The animals were approximately 90 days old at the onset of the experiment. Body weights were from 255 to 380 g for males and 195 to 260 g for females. The animals were housed in standard hanging stainless-steel wire-mesh cages in a vivarium maintained on a 16/8-hr light/dark cycle, with experimental manipulations occurring during the middle portion of the light phase. The animals had free access to food, while water availability was limited to 10 min/day following a progressive deprivation schedule imposed 6 days prior to the start of the study. From the time of weaning until the start of the study, all animals were handled for 30 s three times a week.

Apparatus. Twelve chambers, measuring 24.1 × 12.7 × 20 cm (l × w × h), housed in individual environmental isolation chests were used. Two of the chamber sidewalls and the ceiling were opaque Plexiglas, and the remaining sidewalls were clear Plexiglas. Diffuse illumination was provided in each chamber by a 1.12-W bulb located behind one opaque wall. Chamber floors consisted of parallel stainless-steel rods which were interconnected by NE-2 neon bulbs in series with a 1-Mohm resistor. The output from a high-voltage AC power source could be passed through this circuit to provide a 0.5-mA, 60-Hz, constant-current footshock. On one clear wall of each chamber, there was a 4.4-cm-round, 5-cm-deep, clear Plexiglas cylindrically shaped niche. A water-filled lick tube extended 1 cm into the end of the niche and was present throughout the experiment. An infrared photo emitter and detector was mounted 1 cm in front of the tip of the lick tube such that an animal entering the niche broke the beam of light. An electronic circuit monitored the time that the light beam was interrupted. One CS consisted of a click train produced by a 5/s operation of a 24-VAC relay, operated at 15 VAC, attached to the outside of an opaque wall of each chamber. The click was accompanied by a slight vibration of the chamber walls and floor and was 10 dB(C), re. SPL, above a 70 dB(C) background produced by a ventilation fan mounted on a side wall of each environmental chest. A second CS was provided by a 1500-Hz tone delivered by a 45-ohm overhead speaker. The tone was 12 dB(C) above background.

Procedure. On Days 1 and 2, all animals received one 30-min session/day in the conditioning chamber. These sessions allowed the animals to acclimate to the conditioning enclosures and to establish baseline rates of nosepoking into the niche which contained the lick tube. Latencies to nosepoke for a first and second 5 cumulative s were recorded on each day for all subjects. Animals were randomly assigned to treatment conditions with the restriction that the groups be counterbalanced to the extent possible for bodyweight, litter of origin, sex, and Day-2 latencies to complete Seconds 5-10 in the niche.

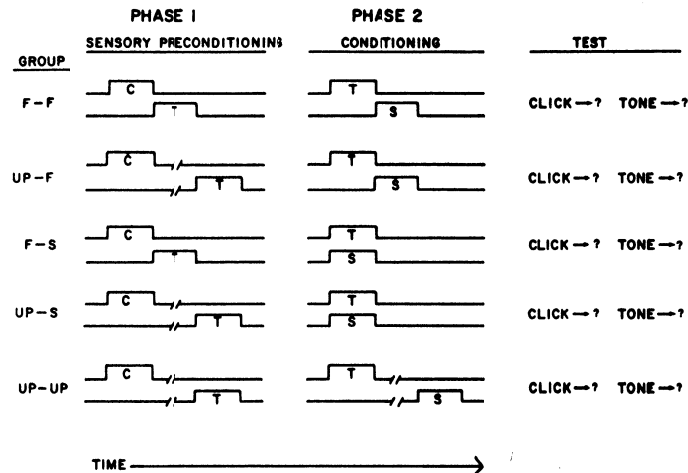


FIG. 1. Schematic depiction of the procedure of Experiment 1. Each click, tone, and shock presentation was 5 s in duration. Each event pair occurred four times in Phase 1 and four times in Phase 2. Has marks (//) indicate a gap in the time line.

Sensory preconditioning (Phase 1) occurred on Day 3. Groups F-F (forward-forward, $n = 20$) and F-S (forward-simultaneous, $n = 19$) received four presentations of a 5-s click which terminated with the onset of a 5-s presentation of the tone. These pairings occurred in a single 60-min session in which the pairings were initiated 10, 20, 37, and 50 min into the session. Groups UP-F, UP-S, and UP-UP (unpaired-forward, unpaired-simultaneous, and unpaired-unpaired, all $n = 19$) received the same number of click and tone presentations, but these stimuli were explicitly unpaired, with clicks occurring at 6, 22, 39, and 51 min, and tones occurring at 11, 16, 34, and 56 min into the session.

Phase 2 conditioning occurred on Day 4. Groups F-S and UP-S received the tone stimulus simultaneously with a 5-s, 0.5-mA footshock. These trials were distributed in the same manner as the sensory preconditioning trials of Phase 1. Groups F-F and UP-F received the same tone presentations, but shock onset coincided with tone termination rather than tone onset; that is, the tone had a forward relationship to the shock. Group UP-UP received the tone and shock unpaired in this phase according to the same unpaired schedule used in the sensory preconditioning phase, except that footshock replaced the clicks.

On Day 5, all animals were placed in the experimental enclosure for 60 min without any discrete stimulus presentations in order to ensure a stable rate of nosepoking. On Day 6, all animals were tested for suppression of ongoing drinking behavior (i.e., nosepoking) in the presence of the clicks, and on Day 7 suppression was tested in the presence of the tone. Testing consisted of placing an animal in its enclosure and allowing it to complete 5 cumulative s in the nosepoke niche. At the completion of the initial 5 s of nosepoking, the appropriate CS was presented and the latency to complete an additional 5 s of nosepoking was recorded. Animals not completing 5 cumulative s in the niche within 60 s of their placement in the chamber were eliminated from the study. All animals remained in the chambers for 10 min after CS onset and a 10-min ceiling was placed on the latency to complete 5 cumulative s in the niche in the presence of the CS. Figure 1 summarizes the procedure used in this study. Day 6 and 7 latencies to complete a first and second 5 s of

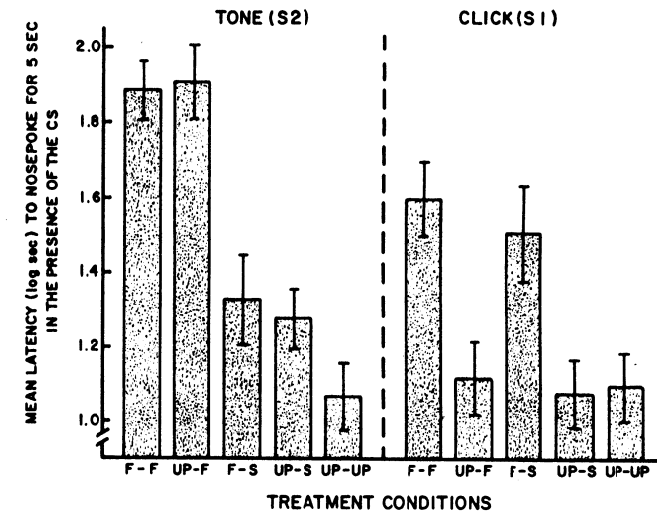


FIG. 2. Mean latencies on the test days to complete 5 cumulative s in the lick-tube niche (nosepoke) in the presence of the clicks (Day 6) and in the presence of the tone (Day 7) in Experiment 1. F, forward pairings of clicks and tone in Phase 1; UP, unpaired presentations of the clicks and tone in Phase 1; -F, forward pairings of tone and shock in Phase 2; -S, simultaneous pairings of tone and shock in Phase 2; -UP, unpaired presentations of tone and shock in Phase 2. Brackets indicate standard errors.

nosepoking in the lick-tube niche were transformed to log latencies to permit the use of parametric statistics.

Results and Discussion

Two subjects (one each from Groups UP-S and UP-UP) were eliminated from the study due to illness and one subject (from Group UP-UP) was eliminated due to failure to complete 5 cumulative s in the niche within 60 s on Day 5. There were no significant between-group differences in latencies to complete an initial 5 s in the niche on either test day, $F_s < 1.00$, and all latencies were less than 1.00 log s.

Figure 2 depicts mean latencies to nosepoke (i.e., suppression of drinking-related behavior) in the presence of the tone (left panel) and clicks (right panel). A 2×2 factorial ANOVA on latencies to nosepoke for 5 s in the presence of the tone (S2) was conducted, with Phase 1 and Phase 2 treatments as factors (Group UP-UP was excluded). The Phase 2 conditioning treatment (forward or simultaneous tone and shock) proved significant, $F(1, 72) = 37.57, p < .001$. Phase 1 sensory preconditioning treatment had no effect on the response to the tone, $F = 0.05$, and Phase 1 and Phase 2 treatments did not interact with respect to the response to the tone, $F = 0.14$. A second ANOVA with treatment as

a single factor was conducted with all five treatment conditions, $F(4, 89) = 15.79, p < .001$. The error term from this analysis was used to conduct planned comparisons. These comparisons indicated that the groups trained with the tone in a forward relationship to the shock (F-F and UP-F) did not differ, $F = 0.01$, and no differences existed between the groups trained with a simultaneous tone and shock (F-S and UP-S), $F = 0.14$. Both of the groups trained with the forward relationships suppressed more than the groups trained with the simultaneous relationships, $F_s(1, 89) \geq 17.57, p_s < .001$, indicating the superiority of forward conditioning to simultaneous conditioning in a direct test of the stimulus paired with the US. Only a tendency toward more suppression in Group F-S than Group UP-UP, $F(1, 89) = 3.98, .05 < p < .10$, and even a weaker tendency toward more suppression in Group UP-S than Group UP-UP, $F(1, 89) = 2.50, .10 < p < .20$, was found, suggesting that simultaneous conditioning may have supported a marginal conditioned response relative to the unpaired control.

Our central interest was in the response to the click stimulus (S1), which predicted the onset of the tone for Groups F-S and F-F. An ANOVA with Phase 1 sensory preconditioning treatment and Phase 2 conditioning treatment as factors was conducted (Group UP-UP was not included in this analysis). The effect of Phase 1 treatment (forward paired or unpaired click and tone) in response to the click proved significant, $F(1, 72) = 17.19, p < .001$. However, Phase 2 treatment (forward or simultaneous conditioning of the tone) had no effect on the response to the clicks, $F = 0.39$, and Phase 1 and Phase 2 treatments did not interact, $F = 0.08$, indicating no superiority of forward over simultaneous conditioning in response to a stimulus that predicts the onset of the forward or simultaneous cue. A single-factor ANOVA comparing the response to the clicks in all five treatment groups found a significant effect of treatment, $F(4, 89) = 5.48, p < .001$. The error term from this analysis was used to conduct planned comparisons of responding to the clicks. Groups F-S and F-F did not differ, $F = 0.35$. Group F-F suppressed more than Group UP-F, $F(1, 89) = 10.08, p < .01$, indicative of a conventional sensory preconditioning effect. That is, the click, which predicted the onset of an excitatory tone, itself acted as an excitor. Similarly, Group F-S suppressed more than Group UP-S, $F(1, 89) = 8.09, p < .01$, although in this case the click predicted the onset of a tone that did not directly elicit a CR due to its simultaneous pairings with the US. None of the groups which received unpaired click and tone presentations in the sensory preconditioning phase (Groups UP-F, UP-S, and UP-UP) differed among themselves in response to the click, $F_s \leq 0.07$.

The current results are consistent with previous work demonstrating that simultaneous conditioning imparts upon a stimulus a potential to

evoke an anticipatory conditioned response, but one that is weak relative to that which results from forward conditioning. Animals that received the tone (S2) paired simultaneously with the shock displayed a weak CR relative to those animals that received the tone-shock forward pairings. Moreover, the tone-shock simultaneous pairings instilled the tone with only marginal response strength relative to animals which received the tone and shock unpaired.

The response to the clicks (S1), which predicted the onset of the tone in Groups F-F and F-S, presents an entirely different picture than that which is typically accepted regarding the efficacy of simultaneous conditioning. The prevailing view has been that simultaneous conditioning produces little or no learning. According to Pavlov (1927) this occurs because of the asynchrony of the neural traces of the CS offset and US onset, and in more recent years explanations have emphasized that a simultaneous CS is uninformative (e.g., Cantor, 1981; Rescorla, 1972, as inferred in Rescorla, 1980b). However, the response to the clicks in the current study was equally strong regardless of whether the clicks predicted the onset of a CS with a simultaneous or forward relationship to the US. This result suggests that simultaneous and forward conditioning support equally robust associations to the US. However, the association formed when the CS and US are paired simultaneously may not be expressed in the traditionally monitored anticipatory response system, presumably because the CS does not evoke an expectation of the US onset. This reasoning, while consistent with the behavioral predictions of the informational viewpoint, is not consistent with the mechanism implied by that theory. Rather than information about a cause and effect relationship determining the formation of the association, mere contiguity between events appears sufficient to support learning. However, information concerning temporal relationships between stimuli appears to determine the nature of the response. In the present instance, a forward CS-US relationship was necessary for the CS to elicit a CR. This implies that an association can incorporate a number of attributes beyond the mere interstimulus associative linkage (e.g., Spear, 1978), one of which is the temporal relationship between the stimuli. Responding to the clicks (S1) was equally robust regardless of whether the tone-shock pairing was simultaneous or forward. This suggests that comparable tone-shock associations were formed in both cases. However, only in the forward case did the tone (S2) provide predictive information concerning the US, thereby directly supporting a conditioned response.

The current paradigm overcomes several problems associated with previous attempts to study simultaneous conditioning. First, the isolated test of a CS that had previously been presented in simultaneous compound with a US results in a dramatic context change between the conditions of training and testing (Rescorla, 1980a). The current procedure overcomes

that possible confound because the click was presented in both training and testing without a stimulus immediately prior to it or concurrent with it. Between groups, the click differed only in whether it predicted a tone, and whether that tone was paired in a forward or simultaneous relation to footshock.

Second, Rescorla suggested that the simultaneous pairing of a CS and a US provides a situation where attention may be diverted from the CS as a consequence of the relatively greater salience of the US; that is, the US might distract the subject from processing of the CS. If this were a viable explanation of the simultaneous conditioning deficit, no response to the clicks would have been elicited in Group F-S since the tone-US association would not have been formed in Phase 2. Since equally strong responding to the clicks was observed in Groups F-S and F-F, information processing models that stress limited capacity at the level of stimulus input (e.g., Broadbent, 1958; Atkinson & Shiffrin, 1968) must be considered inadequate with respect to the present data.

EXPERIMENT 2

Like simultaneous conditioning, backward conditioning has posed problems for associative models based on simple contiguity. Early in training, backward conditioning typically instills a CS with a weak excitatory response potential relative to forward conditioning despite the CS being equally contiguous to the US in the two cases (e.g., Ayres *et al.*, 1987). After extended training, the CS ordinarily proves to be inhibitory (Heth, 1976; Siegal & Domjan, 1971). The transient weak excitation might reflect an associative linkage of the stimuli representations prior to acquisition of knowledge concerning the specific temporal relationship between the stimuli. But the enduring, subsequent conditioned inhibition is more of a problem. It indicates that contiguity is sufficient to support learning, but the nature of what is learned during backward pairings is different from what is learned during forward pairings, despite the same degree of contiguity. The present reasoning suggests that the difference in what is learned in the two cases is the temporal relationship of the two stimuli rather than the degree of learning. Experiment 2 was designed to test the hypothesis that, when a CS and US are paired in a backward as opposed to a forward manner, the principal difference in what is learned is the temporal relationship between the stimuli. Analogous to Experiment 1, the strategy was to see if a stimulus that predicted the onset of a backward-trained CS would act as a conditioned excitor. Specifically, Experiment 2 was a replication of Experiment 1, except that Groups F-B (forward-backward) and UP-B (unpaired-backward) replaced Groups F-S and UP-S. Phase 1 still consisted of click-tone forward pairings for these new groups. But, rather than simultaneous tone-shock presentations

in Phase 2, these new groups received backward pairings of the tone and shock.

Method

Subjects and apparatus. Thirty male and 30 female rats of the same description as those in Experiment 1 served as subjects. The apparatus was identical to that in Experiment 1.

Procedure. Five groups of 12 animals served in this study, which followed the same basic design as Experiment 1. In Phase 1, Groups F-F and F-B received forward click-tone pairings in the sensory preconditioning phase in a manner identical to Experiment 1. Groups UP-F, UP-B, and UP-UP received the clicks and tone unpaired according to the same schedule of unpaired events as in Experiment 1. Phase 2 conditioning proceeded as in Experiment 1. The only difference was that Groups F-B and UP-B received backward pairings of the tone and shock; that is, the 5-s shock terminated with the onset of the 5-s tone. On the day following Phase 2 treatment, all animals spent a 60-min session in the lick chambers without any discrete stimulus presentations in order to ensure stable nose-poking. The subjects were tested for disruption of drinking behavior in the presence of the click on Day 6, and the tone on Day 7. The test procedure was the same as in Experiment 1.

Results and Discussion

Two subjects were eliminated from Group UP-UP prior to testing due to illness. There were no significant between-group differences in latencies to complete an initial 5 s in the niche on either test day, $F_s < 1.00$.

Figure 3 depicts mean latencies to nosepoke (i.e., suppression of drinking behavior) in the presence of the tone (left panel) and clicks (right panel). A 2×2 factorial ANOVA on latencies to nosepoke for 5 s in the presence of the tone (S2) was conducted (Group UP-UP was excluded). The Phase 2 treatment (forward or backward tone-shock presentations) proved significant, $F(1, 44) = 14.78$, $p < .001$. Phase 1 sensory preconditioning treatment had no effect on the response to the tone, $F = 0.03$, and Phase 1 and Phase 2 treatments did not interact, $F = 0.32$.

A second ANOVA with treatments as a single factor was conducted on all five groups. A significant effect of treatment was found, $F(4, 53) = 4.91$, $p < .01$, and the error term from this analysis was used to conduct planned comparisons. These comparisons indicated that the groups trained with the tone-shock forward pairings (F-F and UP-F) did not differ from each other, $F = 0.07$, and neither did the groups trained with the backward arrangement (F-B and UP-B), $F = 0.26$. However, both of the forward groups differed from both of the backward groups as well as the unpaired control group, $F_s(1, 53) \geq 5.74$, $p_s < .05$, indicating a superiority of forward conditioning over backward conditioning in supporting a direct excitatory response to the stimulus that had been paired with the US. Groups F-B and UP-B demonstrated no more suppression to the tone than did the unpaired control group (UP-UP), $F_s \leq 0.30$.

Our primary interest was in the response to the clicks (S1) which predicted the onset of the tone for Groups F-F and F-B. An ANOVA with Phase 1 sensory preconditioning treatment and Phase 2 conditioning

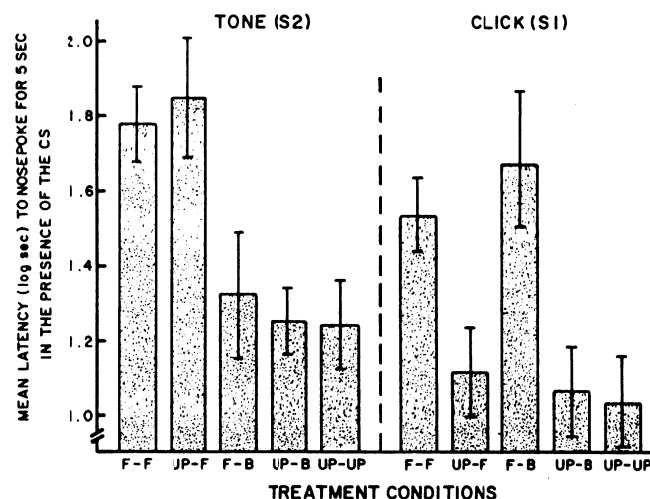


FIG. 3. Mean latencies on the test days to complete 5 cumulative s in the lick-tube niche (nosepoke) in the presence of the clicks (Day 6) and in the presence of the tone (Day 7) in Experiment 2. F, forward pairings of clicks and tone in Phase 1; UP, unpaired presentations of clicks and tone in Phase 1; -F, forward pairings of tone and shock in Phase 2; -B, backward pairings of shock and tone in Phase 2; -UP, unpaired presentations of tone and shock in Phase 2. Brackets indicate standard errors.

treatment as factors was conducted. Group UP-UP was not included in this analysis. The effect of Phase 1 treatment (forward paired or unpaired clicks and tone) on response to the clicks proved significant, $F(1, 44) = 9.45$, $p < .01$. However, Phase 2 treatment (forward or backward conditioning of the tone) had no effect on the response to the clicks, $F = 0.07$, and Phase 1 and Phase 2 treatments did not interact, $F = 0.39$, indicating no superiority of forward over backward first-order conditioning in response to a second-order stimulus that predicts the onset of the first-order cue. A single-factor ANOVA comparing the response to the clicks for all five groups found a significant effect of treatment, $F(4, 53) = 3.39$, $p < .02$. The error term from this analysis was used to conduct planned comparisons of responses to the click. Groups F-B and F-F did not differ, $F = 0.45$. Group F-F suppressed marginally more than UP-F, $F(1, 53) = 3.21$, $p < .08$, and suppressed appreciably more than UP-UP, $F(1, 53) = 4.60$, $p < .05$, indicative of a conventional sensory preconditioning effect. That is, the clicks acted as an excitator as a result of being paired with a neutral stimulus that was later made excitatory. Similarly, Group F-B suppressed more than Group UP-B, $F(1, 53) = 7.46$, $p < .01$, although in this case (Group F-B) the clicks had been paired with the onset of a tone that itself did not evoke an excitatory

CR. None of the groups (UP-F, UP-B, and UP-UP) that received the clicks and tone unpaired in Phase 1 differed from one another, $F_s \geq 0.19$.

Like simultaneous conditioning, backward conditioning is known to instill the CS with a weak excitatory potential relative to forward conditioning (see Spetch, Wilkie, & Pinel, 1981, for a review). The current results suggest that this lack of robust excitatory responding is not due to an absence of underlying associative strength, as is implied by informational accounts of Pavlovian learning. Instead, associative learning seems to be supported by the contiguous occurrence of events regardless of the order of the stimuli. However, the expression of the response elicited by the backward-trained CS appears to be determined in part by the temporal relationship between the cues. In the present instance, the clicks provided information about the onset of a CS trained in a backward relation with the US, and presumably the temporal relationship between the clicks and the US, via its association with the tone, was sufficient to evoke an excitatory response. With appropriate training parameters, backward conditioning is known to produce inhibitory associations. However, the current procedures likely did not instill the CS with inhibitory power owing to the relatively few US-CS pairings that were administered (Heth, 1976; Siegal & Domjan, 1974). In fact, studies administering small numbers of backward pairings ordinarily have yielded weak excitatory responding (Burkhardt, 1980; Heth, 1976; Shurtleff & Ayres, 1981).

The possibility that attributes of the CS and the US beyond the mere associative linkage determine the form of the CR in backward conditioning has been suggested previously by Keith-Lucas and Guttman (1975), who argued that animals should be able to respond to a stimulus perceived as a predatory threat, even if the threat was only noticed *after* an attack. In fact, Keith-Lucas and Guttman observed stronger fear responses to a model of a predator that had previously been presented following footshock than to a simple visual CS that was presented prior to the shock. While these results suggest that a backward CS can indeed evoke an anticipatory response that is stronger than that resulting from forward conditioning, Keith-Lucas and Guttman confounded stimulus characteristics with the direction of conditioning, rendering their results somewhat inconclusive. Moreover, Estes (1979) has argued that a one-trial training procedure such as that used in the Keith-Lucas and Guttman study might be insufficient for encoding temporal order information, thereby resulting in excitatory responding after one trial (or a few) which diminishes after repeated trials (e.g., Heth, 1976). Possibly all instances of excitatory responding to a CS trained with a backward (or simultaneous) relationship to a US represent a failure to adequately encode information concerning temporal order, and might be expected to diminish with further training.

EXPERIMENT 3

The results of Experiments 1 and 2 provided evidence that a forward relationship between two stimuli is important if an association between those stimuli is to be expressed in an anticipatory response system, although a forward relationship did not appear to be necessary for the establishment of the association itself. An alternative to this interpretation has been provided by Wagner (1976). Relative to the first presentation of a novel stimulus, repeated exposure to that stimulus typically attenuates the unconditioned response. Wagner reported evidence that in certain instances the serial presentation of two stimuli in nonreinforced compound reduced the degree to which the unconditioned response to the initial element of the compound was attenuated after repeated exposures. Pfautz, Donegan, and Wagner (1978) have argued that this "protection from habituation" might give rise to the false impression of conditioned responding in a sensory preconditioning paradigm in which animals that receive S1-S2 pairings followed by S2-US pairings are compared to animals that receive S1 and S2 unpaired. This is exactly the procedure employed in Experiments 1 and 2 here. Although Pfautz *et al.* found that the response to S1 after S2-US pairings with this procedure was greater than could be accounted for by protection from habituation, it is possible that this mechanism contributed to the response to S1 observed here, and might therefore account for the equality of the response to S1 regardless of forward, simultaneous, or backward pairings of S2 and the US. In Experiment 3 we tested this possibility by administering forward S1-S2 pairings to three groups of animals in Phase 1, followed by simultaneous, backward, or unpaired presentations of S2 and the US in Phase 2. Enhanced responding in animals that receive simultaneous (as in Experiment 1) or backward (as in Experiment 2) presentations of S2 and the US relative to animals that receive S2 and the US unpaired (not previously run) could not be attributed to protection from habituation.

Method

Subjects and apparatus. Eighteen male and 18 female rats of the same general description as those in Experiment 1 served in this study. The apparatus was identical to that of Experiment 1.

Procedure. As in Experiment 1, all animals received two 30-min adaptation sessions during which they were allowed to drink from the lick tubes on Days 1 and 2. On Day 3 (Phase 1), sensory preconditioning was conducted in the manner described for Group F-S in Experiment 1, that is, all animals received four click-tone forward pairings in a 60-min session. On Day 4 (Phase 2), animals received one of three conditioning treatments in a 60-min session. For Group F-S, treatment was identical to the same named group in Experiment 1. Specifically, this group experienced four simultaneous presentations of the 5-s tone and the 0.5-mA footshock. Group F-B also received four tone and shock presentations, but in this case the offset of the shock was immediately followed by the onset of the tone. That is, these animals received backward pairings of the tone and shock just as had Group F-B in Experiment 2. The final group, Group F-UP received the shock and tone explicitly

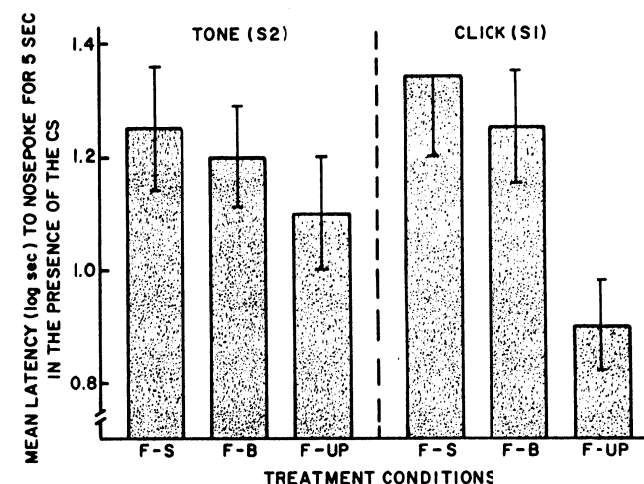


FIG. 4. Mean latencies on the test days to complete 5 cumulative s in the lick-tube niche (nosepoke) in the presence of the clicks (Day 6) and in the presence of the tone (Day 7) in Experiment 3. F, forward pairings of clicks and tone in Phase 1; -S, simultaneous pairings of tone and shock in Phase 2; -B, backward pairings of shock and tone in Phase 2; -UP, unpaired presentations of tone and shock in Phase 2. Brackets indicate standard errors.

unpaired, using the same procedure described for animals receiving unpaired shock and tone in the previous experiments. Lick tubes were present throughout training.

On Day 5, all animals spent a 60-min session in the conditioning chamber without any CS or US presentations in order to ensure stable nosepoking in the lick-tube niche. All animals were tested for suppression of nosepoking in the presence of the click on Day 6, and in the presence of the tone on Day 7. The test procedure was the same as in Experiment 1.

Results and Discussion

Group mean suppression in the presence of the CSs at the time of testing are illustrated in Fig. 4. There were no significant differences in latencies to complete an initial 5 s in the lick-tube niche on either test day, $F_s < 1.0$.

The response to the tone (S2) did not differ as a function of whether the tone was simultaneously paired, backward paired, or unpaired with the shock, $F(2, 33) = 1.04$, indicative of an absence of appreciable conditioned responding to the tone following simultaneous or backward conditioning. Due to an apparatus malfunction, one animal in Group F-UP was not tested with the click. Responding to the click (S1) did differ as a function of the relationship of the tone to the shock, $F(2, 32) = 4.21$, $p < .05$. Group F-UP exhibited less suppression of nosepoking

than either Group F-S, $F(1, 32) = 7.59, p < .01$, or Group F-B, $F(1, 32) = 4.74, p < .05$.

The results of Experiment 3 indicate that the mere sequential pairing of the click and tone does not endow the click with the potential to disrupt responding if the tone is not subsequently paired with the shock US. Like Experiments 1 and 2, sequential pairing of the click and tone did allow the click to disrupt icking if the tone and shock were subsequently paired in a simultaneous or backward manner. Therefore, with these parameters, protection from habituation, or any other process specific to the sequential pairing of the click and tone, appears insufficient to account for the observed responding to the click.

EXPERIMENT 4

Rizley and Rescorla (1972) provided evidence that responding following sensory preconditioning (i.e., S1-S2, S2-US) was mediated by a direct linkage between S1 and S2. Following sensory preconditioning (S1-S2) and S2-US pairings, Rizley and Rescorla extinguished the CR to S2. No evidence of excitation was noted in a subsequent test of S1. Rizley and Rescorla concluded that S1 acts as an excitor by reactivating the association between S1 and S2, provided that S2 evokes a CR, either by a direct S2-CR association or an S2-US association. This observation raises questions concerning the nature of the associative linkage that allows S1 to act as an excitor when S2 itself elicits no overt CR, as in the present experiments. Several possibilities can be excluded. First, S1 does not appear to simply reactivate a representation of S2, with S2 in turn eliciting either a response (see path b of Fig. 5) or reactivating a representation of the US (see path d of Fig. 5). In both of these cases, presentation of S1 should result in a response that is appropriate for S2, that is, no overt response. Moreover, the possibility that S1 is directly associated with the response to S2 without mediation by a representation of S2 (see path a of Fig. 5) does not appear viable. This alternative can be rejected because it too requires that S1 evoke the same response as S2.

There are several remaining interpretations of the present data. The first is that an S1-US association is formed at the time of S2-US pairings (see path c of Fig. 5). Such a linkage would be insensitive to subsequent changes in the associative strength of S2, as S1 would directly reactivate the representation of the US. Moreover, S1 could possibly reactivate not only a representation of S2, but also information about the temporal relationship between S1 and S2. This latter case is consistent with Rizley and Rescorla's conclusions, but implies a somewhat different type of association than the simple stimulus-stimulus linkage which they considered. This latter possibility is encompassed by paths b and d of Fig.

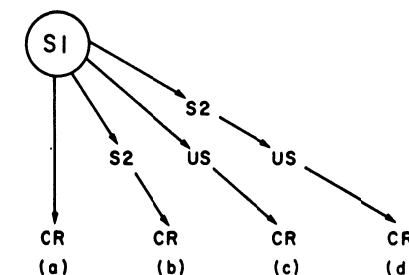


FIG. 5. Schematic depiction of the possible associative linkages whereby the clicks (S1) acquire associative strength as a function of click-tone (S2) pairings in Phase 1 followed by tone-shock pairings in Phase 2.

5, although the figure does not make reference to the putative temporal attributes which may be embedded in the links.

Experiment 4 was intended to delineate the associative linkage that endows S1 with response strength, even when its known associate, S2, does not itself elicit a response. This was accomplished using two techniques, postconditioning revaluation of S2 and postconditioning revaluation of the US. As in Experiment 1, the clicks (S1) and tone (S2) in Phase 1 were paired in a forward relationship. In Phase 2, four groups of animals received tone-shock forward pairings, and four groups received tone and shock simultaneous pairings. Critical treatments were conducted in Phase 3. One forward group (Fnt) and one simultaneous group (Snt) received no further treatment. One forward group (Fet) and one simultaneous group (Set) received extinction trials with the tone. Such trials would be expected to disrupt the associative linkage supporting the CR in paths b and d of Fig. 5, as has been demonstrated for forward-forward sensory preconditioning by Rizley and Rescorla (1972). In addition, one forward group (F+) and one simultaneous group (S+) received (in an irrelevant context) shock presentations of a higher intensity (US inflation) than that used in initial conditioning in order to inflate the representation of the US. This inflation technique has been shown to enhance the CR to a CS previously paired in a forward manner with a US (Rescorla, 1974). Any changes in the magnitude of the present CR as a result of this inflation manipulation would be expected to be manifestations of the linkages represented by paths c and d in Fig. 5. Collectively, extinction of the tone and inflation of the value of the shock should allow us to identify the associative linkage responsible for the CR to the clicks (S1).

Method

Subjects and apparatus. Twenty-seven male and 27 female rats of the same description as those in Experiment 1 served in this study. The apparatus was identical to that of

Experiment 1 except for the addition of 12 copies of a second enclosure in which noncontingent shock could be administered in Phase 3. This added enclosure consisted of a $50 \times 8.2 \times 30$ -cm ($l \times w \times h$) stall with opaque green Plexiglas walls and floors composed of 0.64-cm-diameter electrifiable grids (spaced 1.59 cm center-to-center) interconnected by NE-2H neon bulbs. There were no lick tubes present. These enclosures were housed in a large room that was brightly illuminated by overhead fluorescent lights. The ambient sound level was 56 dB(C).

Procedure. As in the previous experiments, all animals received two 30-min sessions of adaptation to the conditioning chamber prior to any treatment. On Day 3 (Phase 1) of the study, all animals received four click-tone pairings in the same forward manner as the paired animals in Phase 1 of the prior studies. The animals were then divided into six groups of nine rats each for the second phase of the study. On Day 4 (Phase 2), Groups Fnt, Fet, and F+ received four forward (F) tone-shock pairings on Day 4, and Groups Snt, Set, and S+ received four simultaneous (S) tone and shock presentations. Tone and shock presentations were scheduled as in the previous studies. Phase 3 was conducted on Days 5 and 6, during which Groups Fet and Set received eight daily 5-s presentations of the tone alone in the conditioning chamber, randomly distributed in a daily 60-min session. These presentations were intended to extinguish (et) any CR to the tone. During these same 2 days, Groups F+ and S+ were placed in the noncontingent shock chambers where they received four 1.2-mA, 5-s unsignaled shocks (+) in each of two 60-min sessions distributed according to the same schedule as in initial conditioning. These inflation shocks were intended to enhance the US representations for Groups F+ and S+. During this 2-day period, Groups Fnt and Snt spent two 60-min sessions in the conditioning chambers without CS or US presentations, that is, no treatment (nt).

On Day 7 all animals spent a 60-min session in the conditioning chamber without any CS or US presentations in order to ensure stable nosepoking (drinking) in the lick-tube niche. All animals were tested for suppression of nosepoking in the presence of the clicks on Day 8, and in the presence of the tone on Day 9.

Results and Discussion

There were no significant between-group differences in latencies to complete an initial 5 s in the niche on either test day, $F_s < 1$.

Group mean suppression in the presence of the CSs at the time of testing is illustrated in Fig. 6. A 2×3 factorial ANOVA on latencies to nosepoke for 5 s in the presence of the tone (S2) was conducted. Significant effects of Phase 2 treatment (simultaneous vs forward conditioning), $F(1, 48) = 9.73, p < .01$, Phase 3 treatments (no treatment, extinguish tone, or inflate shock), $F(2, 48) = 34.99, p < .001$, and the interaction of Phases 2 and 3 treatment, $F(2, 48) = 4.66, p < .05$, were obtained. Planned comparisons found that operational extinction of the tone reduced the CR to the tone in Group Fet relative to Group Fnt, $F(1, 48) = 21.01, p < .001$. As would be expected due to the absence of a CR to the tone in Group Snt, no such reduction in response to the tone was seen in Group Set relative to Group Snt, $F = 0.09$. The effect of postconditioning shock inflation on the response to the tone was somewhat surprising. For animals that received forward tone-shock training in Phase 2, inflation of the shock after conditioning enhanced the response to the tone in Group F+ relative to Group Fnt, $F(1, 48) = 4.76, p <$

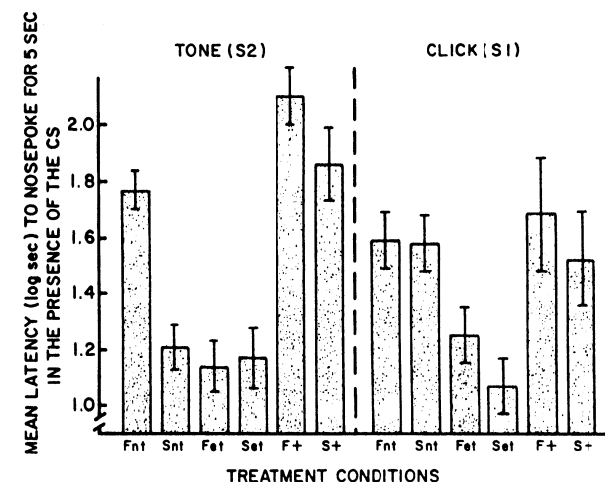


FIG. 6. Mean latencies on the test days to complete 5 cumulative s in the lick-tube niche (nosepoke) in the presence of the clicks and tone in Experiment 4. In Phase 1, all subjects received click-tone forward pairings. In Phase 2, tone was paired with footshock either in a forward (F) or simultaneous (S) manner. In Phase 3, either the tone was extinguished (et), the shock representation was inflated (+) through delivery of high-intensity shock, or no treatment (nt) was given. Brackets indicate standard errors.

.05. Groups trained with simultaneous tone and shock presentations in Phase 2 were not expected to exhibit any response changes as a result of inflation of the shock since the simultaneous trained cue did not elicit an excitatory CR. However, this was not the case. Relative to Group Snt, inflation of the shock enhanced the response to the tone, $F(1, 48) = 22.36, p < .001$. It is not clear at the present time what the mechanism of this response enhancement might be (see below).

The central concern of the present study was to determine what effect, if any, postconditioning alteration of the response evoking potential of the tone would have on responding to the clicks, which anticipates the onset of the tone. A two-factor ANOVA found no reliable effect of Phase 2 (simultaneous or forward training) treatment on the response to the click, $F(1, 48) = 1.06, p > .25$. Phase 3 treatments did prove significant, $F(2, 48) = 6.57, p < .01$, but no interaction was observed between Phase 2 and Phase 3, $F = 0.20$. Planned comparisons found that Group Fet suppressed marginally less than Group Fnt, $F(1, 48) = 3.26, p < .10$, suggesting that extinction of the tone attenuated the response to the clicks in animals trained with a forward tone-shock arrangement. More important to the current analysis, extinction of the tone eliminated responding to the clicks in Group Set relative to Group Snt, $F(1, 48) = 7.31, p < .01$. Postconditioning inflation of the footshock had no effect

on the response to the clicks in either the group trained with forward tone-shock pairings, $F = 0.10$, or the group trained with simultaneous tone-shock pairings, $F = 0.14$, relative to their appropriate no-treatment controls.

The current results implicate the click-tone (S1-S2) associative link in mediating the response to the clicks in the case of simultaneous conditioning. Extinction of the tone after simultaneous presentations of the tone and shock reduced responding to the clicks to baseline levels (see Fig. 6). This finding eliminates paths a and c in Fig. 5 as possible associative links responsible for responding to the clicks after simultaneous tone-shock pairings in Phase 2, as the tone is not represented in either of these linkages. (It should be noted that generalized extinction could account for the loss of responding to the click when the tone was extinguished after forward tone-shock pairings, but not after simultaneous tone-shock pairings, since in the latter case the tone did not evoke a conditioned response prior to extinction. The possibility of generalized extinction can be further discounted in forward as well as simultaneous conditioning based on prior work by Matzel, Schachtman, and Miller (1985, Experiment 3), who, using similar stimuli and identical apparatus to study overshadowing, found no evidence of generalization of extinction between the click and tone.)

Although postconditioning inflation of the shock representation increased responding to the tone trained simultaneously with shock, this manipulation had no effect on the response to the click, which predicted the onset of the tone since a representation of the tone itself (reactivated by presentation of the tone) is not sufficient to elicit a response. This suggests that the associative linkage represented by path b in Fig. 5 was mediating the response to the click. In total, these results support the conclusion of Experiment 1 that the presentation of the click evokes not only a representation of the tone, but also information regarding the temporal relationship between the clicks and the tone. (A more direct test of this hypothesis is conducted in Experiment 5.)

Forward conditioning groups were included in the current study in order to evaluate the effectiveness of the Phase 3 treatments, as we did not anticipate that these treatments would be directly manifest in the response to the simultaneous trained tone (S2). However, inflation of the shock after simultaneous conditioning with the tone did in fact *enhance* the CR to the tone. One possible account of this effect can be based on generalized fear (Pavlov, 1927) resulting from Phase 3 shock-alone presentations. However, this is unlikely for two reasons. First, the shock-alone presentations occurred outside of the test context, which would be expected to minimize any transfer of fear to the test context. Second, while inflation of the US value increased the CR to the tone (S2) after both the forward and simultaneous tone-shock pairings, this inflation of

the shock had no effect on the response to the click. Unless we assume that the response to the tone is differentially affected by generalized fear, generalized fear does not seem to be a viable explanation of these results.

A second possible mechanism underlying the enhanced responding that was observed following simultaneous conditioning and shock inflation is that the shock acts as a "reminder" of the tone-US association. At least in the case of backward conditioning, such treatments have been found to enhance responding (Gordon, McGinnis, & Weaver, 1985). In the current case, however, it is not clear of what the animal is reminded, as Experiments 1 and 4 both suggest that the association formed in simultaneous conditioning is as good as the association formed in forward training, but differs in the form of the response that the CS will support. Possibly, the shock-alone presentations "confused" the animal regarding the temporal order of the tone and shock. A similar explanation has been applied by Estes (1979) to explain the excitation often seen to a simultaneous trained CS after a small number of trials, that is, before the animal has acquired sufficient information to appreciate the temporal nature of the associative linkage.

EXPERIMENT 5

Experiment 5 was intended as a further test of the possibility, suggested by the previous experiments, that a forward relationship between a CS and US facilitates the expression of an association in an anticipatory response system, but is not necessary for the establishment of the association. The principal conclusions of Experiments 1-4 were based on the assumption that a forward relationship in Phase 1 between the clicks and tone was important if the clicks were to support a CR following simultaneous tone-shock pairings. Rescorla (1980a) in fact has demonstrated strong sensory preconditioning when the Phase 1 (S1-S2) pairings were simultaneous and Phase 2 (S2-US) pairings were forward. Like Experiments 1-4 here, Rescorla's preparation provided S1 with a forward relationship to the US via its association with S2. If a forward temporal relationship between a CS and US facilitates the expression of an anticipatory response, then simultaneous S1-S2 pairings followed by simultaneous S2-US training should not instill S1 with response-evoking properties. In Experiment 5, three groups of animals (S-S, S-F, and S-UP) received simultaneous S1-S2 training in Phase 1. In Phase 2, Group S-F received forward S2-US pairings, Group S-S received simultaneous S2-US pairings, and Group S-UP received S2 and the US explicitly unpaired. The response evoked by S1 was of primary interest. If a forward relationship promotes the expression of learning in anticipatory response systems, then S1 should evoke a conditioned response only in Group S-F.

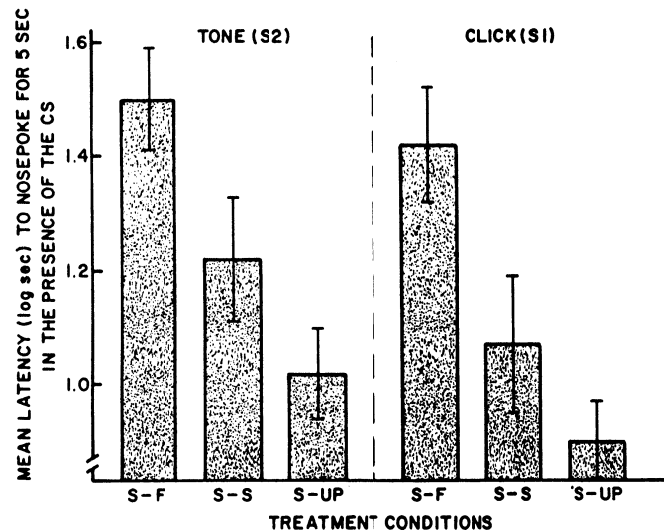


FIG. 7. Mean latencies to complete 5 cumulative s in the lick-tube niche (nosepoke) in the presence of the clicks and tone in Experiment 5. The clicks and tone were paired simultaneously (S) in Phase 1 and the tone and shock were presented in a forward (-F), simultaneous (-S), or unpaired (-UP) manner in Phase 2. Brackets indicate standard errors.

Method

Subjects and apparatus. Eighteen male and 18 female rats of the same general description as in Experiment 1 served in this study. Twelve animals were assigned to each group. The apparatus was the same as that in Experiment 1.

Procedure. All animals were adapted to the experimental enclosures on Days 1 and 2. On Day 3, each animal received four simultaneous presentations of the 5-s click and tone distributed in a 60-min session according to the schedule described in Experiment 1. During a 60-min session on Day 4 (Phase 2), Group S-S received four simultaneous tone-shock presentations, Group S-F received four forward tone-shock presentations, and Group S-UP received the tone and shock unpaired. These events were scheduled just as the corresponding treatments in Experiment 1.

All animals received a 60-min session in the conditioning chamber on Day 5 without any discrete stimulus presentations in order to ensure stable rates of nosepoking. Testing (i.e., suppression of nosepoking in the lick-tube niche in the presence of the CS) occurred first with the click on Day 6, and then with the tone on Day 7, and was conducted in the manner described in Experiment 1.

Results and Discussion

There were no significant between-group differences to complete an initial 5 s in the niche on either test day, $F_s < 1$.

Figure 7 illustrates the results of this experiment. An ANOVA on latencies to nosepoke for 5 s in the presence of the tone (S2), which for

different groups was forward paired, simultaneously paired, or unpaired with the shock in Phase 2, indicated a significant effect of treatment, $F(2, 33) = 6.26$, $p < .01$. Planned comparisons revealed that Group S-F suppressed more than Group S-UP, $F(1, 33) = 12.80$, $p < .01$, whereas Groups S-S and S-UP did not differ, $F(1, 33) = 2.22$. Group S-F suppressed only marginally more than Group S-S, $F(1, 33) = 4.35$, $p < .07$. This latter comparison alone might be construed to suggest that the CR to the forward-trained CS was no stronger than that to the simultaneous trained CS. However, the results of this study as well as other experiments reported here collectively suggest that forward tone-shock pairings resulted in a tone that evoked substantially stronger responding than did simultaneous tone-shock pairings.

The response to the clicks (S1), which had been simultaneously paired with the tone in Phase 1, closely paralleled the response to the tone. An ANOVA indicated a significant effect of treatment, $F(2, 33) = 7.57$, $p < .01$. Groups S-S and S-UP did not differ, $F(1, 33) = 1.38$, while Group S-F suppressed more than Group S-S, $F(1, 33) = 7.00$, $p < .02$, and Group S-UP, $F(1, 33) = 14.61$, $p < .01$.

The results of this experiment indicate that a forward temporal relationship between a CS and US facilitates the expression of an anticipatory CR to the CS. In combination with Experiment 1, these observations indicate that in a sensory preconditioning preparation either a forward relationship between S1 and S2, or S2 and the US, is required if S1 is to evoke an anticipatory response appropriate for that US. However, it is sufficient for either of these associative connections to be forward. Either the simultaneous pairing of S1 with S2 or S2 with the US effectively supported learning and, provided S1 had an anticipatory relation to the US, endowed S1 with the ability to evoke a response which reflected information concerning the impending US.

GENERAL DISCUSSION

The principal finding of the present experiments was that following simultaneous (Experiment 1) or backward (Experiment 2) pairings of a tone and footshock, a click stimulus that predicted the onset of the tone acted as a conditioned excitor. This occurred despite the fact that the tone itself did not elicit a CR. The first and most obvious implication of these results is that the response elicited by a CS is not necessarily indicative of the underlying associative strength of that stimulus, but may also depend on the appropriateness of the response given the temporal relationship of the CS and US (also see Blanchard & Honig, 1976; Rescorla, 1980a). The tone in the present experiments did not elicit a CR, presumably because simultaneous or backward pairings of the tone and shock instilled the tone with no predictive value. However, the response to the clicks,

which anticipated the onset of the tone, indicates that the tone was associated with the shock.

A second implication of the current results is that this type of predictive information, although important for an overt conditioned response, is not necessary for the establishment of an association. However, if the response that we monitor is anticipatory in nature (e.g., fear of an upcoming shock), then it is not surprising that this response does not reflect learning in the case of simultaneous or backward conditioning. In both cases, the CS has no predictive relationship regarding the shock. This suggests that the informational view of learning is inherently flawed and might better be replaced by an informational view of responding.

Experiment 4 determined that the response to the clicks (S1) was sensitive to the extinction of the tone (S2). This finding implies that the associative linkage that provides the clicks with response strength is one in which the clicks reactivate a representation of the tone including the temporal relationship of the tone to the clicks. Such a conclusion concerning the temporal attributes of the S1-S2 association was strengthened by the results of Experiment 5 in which it was determined that either the click-tone or tone-shock pairings must be sequential if the clicks are to elicit a CR. Simultaneous click-tone pairings followed by simultaneous tone-shock pairings instilled the click with no response potential.

The suggestion that the form of the CR is a function of the temporal relationship between the CS and US and consequently depends on the temporal appropriateness of the particular response system being monitored is not without precedent. Farley and Alkon (1987) as well as Grover and Farley (1987) have repeatedly demonstrated a superiority of simultaneous to forward conditioning. While this observation has been made with *Hermisenda* as subjects, it is likely due to the response system in question rather than the species studied. These researchers monitored depolarization of photoreceptors in response to a light CS that had previously been paired with a US (rotation). Since the stimulus receptor in this case is also the generator of the response, direct responding is evidenced, as opposed to anticipatory responses (e.g., eyeblink, suppression, salivation) which are several steps removed from the underlying substrates of the association. However, in at least one instance, the *behavioral* response (suppression of phototaxis) of *Hermisenda* elicited by a CS was greater when the CS was previously trained with a forward relationship to the US than when the two stimuli had been presented simultaneously (for discussion, see Matzel, Schreurs, Lederhendler, & Alkon, 1988).

That evidence of an association apparently depends on the particular response system being observed is currently a popular notion in the human learning literature. For instance, Perruchet (1985), using human subjects, demonstrated that a single CS-US association trial could have

negatively correlated effects upon two different response systems used to measure the strength of the association (in this case eyeblink or a verbal report of expectation of the US). Similarly, Jacoby (1983) has found that the translation of memory to performance is dependent on the nature of the task, and seemingly opposing results can be attained depending on the choice of response system.

Results similar to those presented here have been provided by Heth (1976). Heth reported that a backward-trained CS, while not capable of eliciting excitatory conditioned responding, was capable of punishing and hence suppressing barpressing for appetitive reinforcement when the barpress produced the CS. He interpreted these results as suggesting that the CS had acquired weak excitatory strength. However, his results may reflect a second-order conditioning analog to the current experiments. Following backward pairings of a CS with shock, a barpress that results in the onset of the CS *anticipates* the onset of that CS and should function as a good predictor of shock, thereby suppressing barpressing. While Heth's data are entirely compatible with this analysis, the alternative interpretation that backward conditioning produces a weak association is not compatible with the results of Experiment 2 here. Backward associations were indistinguishable from forward associations when a stimulus that predicted the onset of the backward- or forward-trained CS provided the index of response strength.

Rescorla (1980a) hypothesized that the failure of simultaneous conditioning to support learning may result from interference by the US with the processing of the CS at the time of initial conditioning. In order to test this possibility, Rescorla paired two neutral stimuli (S1-S2), in either a forward or a successive manner, and subsequently forward-paired S2 with a US. In a test of S1, Rescorla found that simultaneous training with S2 instilled S1 with greater response strength than did the forward pairings. However, Rescorla (1981) concluded that the response to the simultaneous trained S1 was likely not the result of associative learning, but rather, probably reflected perceptual configuring, as preexposure to either S1 or S2 alone disrupted conditioning. This conclusion left unresolved the issue of whether simultaneous conditioning was sufficient to support learning. However, the results of the experiments reported here cannot be readily explained in terms of configuring. If perceptual configuring of the clicks and tone had occurred during the sensory pre-conditioning phase, the clicks would be expected to elicit the response appropriate for the tone, that is, the clicks and tone would act as the same stimulus. This, however, was not the case. The clicks acted as an excitator even when the tone with which it had been paired did not itself evoke a response. Although the configuring hypothesis advanced by Rescorla cannot account for these data, the anticipatory response rule suggested here can explain Rescorla's results. Along similar lines, the

present data argue against stimulus generalization accounts of sensory preconditioning (e.g., Pearce, 1987), as such a mechanism would require that the clicks and tone in the present studies elicit the same response.

It should be noted that an analysis of associative learning based exclusively upon contiguity would predict *better* learning as a result of simultaneous CS-US pairings than forward CS-US pairings (Pavlov, 1927). Neither Experiments 1 or 4 in the current series, which directly compared forward and simultaneous conditioning, found evidence in support of this prediction. However, it is quite possible that the current parameters were not sensitive to such an effect. The relatively short interval between CS onset and US onset (5 s) would be expected to produce a high degree of responding in a conditioned suppression preparation. This could have resulted in a "ceiling" effect that could have obscured any differences in underlying associative strength between forward and simultaneously paired CSs.

Finally, the current results are consistent with numerous previous reports attesting to the sufficiency of contiguity to support learning (Hearst, 1987; Matzel, Brown, & Miller, 1987; Matzel *et al.*, 1985; Miller, Kasprow, & Schachtman, 1986; Timberlake, 1986), and are contrary to the view that contiguity alone will not support an association between stimuli (Rescorla, 1988). Moreover, consistent with Matzel *et al.* (1987), we see that caution must be used when applying the term *failure* to learning or performance. The experiments reported here are incompatible with previous conclusions that the relatively weak responding seen after simultaneous and/or backward conditioning reflects a learning deficit (for review, see Moore & Gormezano, 1977). Nor can they be construed as performance failures in the traditional sense, which implies some sort of deficit. The animals in these experiments demonstrated clear evidence of learning simultaneous and backward associations. However, they responded in a way reflecting that information when and only when the information was of functional, predictive value appropriate for the response system that we monitored.

REFERENCES

- Atkinson, R. C., & Saffrin, R. M. (1968). Human memory: A proposed system and its control processes. In K. W. Spence & J. T. Spence (Eds.), *The psychology of learning and motivation: Advances in theory and research* (Vol. 2, pp. 89-195). New York: Academic Press.
- Ayres, J. J. B., Haddad, C., & Albert, M. (1987). One-trial excitatory backward conditioning assessed by conditioned suppression of licking in rats: Concurrent observations of lick suppression and defensive behaviors. *Animal Learning and Behavior*, 15, 212-217.
- Blanchard, R., & Hong, W. K. (1976). Surprise value of food determines its effectiveness as a reinforcer. *Journal of Experimental Psychology: Animal Behavior Processes*, 2, 67-74.
- Broadbent, D. E. (1958). *Perception and communication*. London: Pergamon.
- Brogden, W. J. (1939). Sensory pre-conditioning. *Journal of Experimental Psychology*, 25, 323-332.
- Burkhardt, P. E. (1980). One-trial backward fear conditioning in the rat as a function of US intensity. *Bulletin of the Psychonomic Society*, 15, 9-11.
- Burkhardt, P. E., & Ayres, J. J. B. (1978). CS and US duration effects in one-trial simultaneous fear conditioning as assessed by conditioned suppression of licking in rats. *Animal Learning and Behavior*, 6, 225-230.
- Cantor, M. B. (1981). Information theory: A solution to two big problems in the analysis of behavior. In P. Harzem & M. D. Zeiler (Eds.), *Predictability, correlation, and contiguity* (pp. 287-320). Chichester, UK: Wiley.
- Egger, M. D., & Miller, N. E. (1962). Secondary reinforcement in rats as a function of information value and reliability of the stimulus. *Journal of Experimental Psychology*, 64, 97-104.
- Estes, W. K. (1979). Cognitive processes in conditioning. In A. Dickinson & R. A. Boakes (Eds.), *Mechanisms of learning and motivation: A memorial volume to Jerzy Konorski* (pp. 417-441). Hillsdale, NJ: Erlbaum.
- Farley, J., & Alkon, D. L. (1987). In vitro associative conditioning of *Hermissenda*: Cumulative depolarization of Type B Photoreceptors and short-term associative behavioral changes. *Journal of Neurophysiology*, 57, 1639-1668.
- Gibson, E. J. (1940). A systematic application of the concepts of generalization and differentiation to verbal learning. *Psychological Review*, 47, 196-229.
- Gordon, W. C., McGinnis, C. M., & Weaver, M. S. (1985). The effect of cuing after backward conditioning trials. *Learning and Motivation*, 16, 444-463.
- Grover, L. M., & Farley, J. (1987). Temporal order sensitivity of associative neural and behavioral changes in *Hermissenda*. *Behavioral Neuroscience*, 101, 558-675.
- Hawkins, R. R., & Kandel, E. R. (1984). Is there a cell alphabet for simple forms of learning? *Psychological Review*, 91, 375-391.
- Hearst, E. (1987). Extinction reveals stimulus control: Latent learning of feature-negative discriminations in pigeons. *Journal of Experimental Psychology: Animal Behavior Processes*, 13, 52-64.
- Heth, C. D. (1976). Simultaneous and backwards fear conditioning as a function of number of CS-US pairings. *Journal of Experimental Psychology: Animal Behavior Processes*, 2, 117-129.
- Holland, P. C. (1984). Origins of behavior in Pavlovian conditioning. In G. H. Bower (Ed.), *The psychology of learning and motivation*. New York: Academic Press.
- Jacoby, L. L. (1983). Remembering the data: Analyzing interactive processes in reading. *Journal of Verbal Learning and Verbal Behavior*, 22, 485-508.
- Kamin, L. J. (1969). Predictability, surprise, attention, and conditioning. In B. A. Cambell & R. M. Church (Eds.), *Punishment and aversive behavior* (pp. 279-296). New York: Appleton-Century-Crofts.
- Kandel, E. R., & Schwartz, J. H. (1982). Molecular biology of learning: Modulation of transmitter release. *Science*, 218, 433-443.
- Keith-Lucas, T., & Guttman, N. (1975). Robust single-trial delayed backward conditioning. *Journal of Comparative and Physiological Psychology*, 88, 468-476.
- Mackintosh, N. J. (1983). *Conditioning and associative learning*. Oxford, UK: Oxford Univ. Press.
- Mahoney, W. J., & Ayres, J. J. B. (1976). One-trial simultaneous and backward fear conditioning as reflected in conditioned suppression of licking in rats. *Animal Learning and Behavior*, 4, 357-362.
- Matzel, L. D., Brown, A. M., & Miller, R. R. (1987). Associative effects of US preexposure: Modulation of conditioned responding by an excitatory training context. *Journal of Experimental Psychology: Animal Behavior Processes*, 13, 65-72.

- Matzel, L. D., Schachtman, T. R., & Miller, R. R. (1985). Recovery of an overshadowed association achieved by extinction of the overshadowing stimulus. *Learning and Motivation*, 16, 398-412.
- Matzel, L. D., Schreurs, B. G., Lederhendler, I. I., & Alkon, D. L. (1988). *Acquisition of conditioned associations in Hermisenda: Synergistic effects of the forward inter-stimulus interval and CS-US contiguity*. Manuscript submitted for review.
- Miller, R. R., Kasprow, W. J., & Schachtman, T. R. (1986). Retrieval variability: Sources and consequences. *American Journal of Psychology*, 99, 145-218.
- Moore, J. W., & Gornezano, I. (1977). Classical conditioning. In M. H. Marx & M. E. Bunch (Eds.), *Fundamentals and applications of learning*. New York: MacMillan Co.
- Pavlov, I. P. (1927). *Conditioned reflexes*. London: Oxford Univ. Press.
- Pearce, J. M. (1987). A model for stimulus generalization in Pavlovian conditioning. *Psychological Review*, 94, 61-73.
- Pearce, J. M., & Hall, G. (1980). A model for Pavlovian conditioning: Variations in the effectiveness of conditioned but not unconditioned stimuli. *Psychological Review*, 87, 332-352.
- Perkins, C. C. (1968). An analysis of the concept of reinforcement. *Psychological Review*, 75, 155-172.
- Perruchet, P. (1985). A pitfall for the expectancy theory of human eyelid conditioning. *Pavlovian Journal of Biological Science*, 20, 163-170.
- Pfautz, P. L., Donegan, N. H., & Wagner, A. R. (1978). Sensory preconditioning versus protection from habituation. *Journal of Experimental Psychology: Animal Behavior Processes*, 4, 286-295.
- Prewitt, E. P. (1967). Number of preconditioning trials in sensory preconditioning using CER training. *Journal of Comparative and Physiological Psychology*, 64, 360-362.
- Randich, A., & LoLordo, V. M. (1979). Preconditioning exposure to the unconditioned stimulus affects the acquisition of a conditioned emotional response. *Learning and Motivation*, 10, 245-277.
- Rescorla, R. A. (1972). Informational variables in Pavlovian conditioning. In G. H. Bower (Ed.), *The psychology of learning and motivation* (Vol. 6, pp. 1-46). New York: Academic Press.
- Rescorla, P. A. (1974). Effect of inflation of the unconditioned stimulus value following conditioning. *Journal of Comparative and Physiological Psychology*, 86, 101-106.
- Rescorla, R. A. (1980a). Simultaneous and successive associations in sensory preconditioning. *Journal of Experimental Psychology: Animal Behavior Processes*, 6, 207-216.
- Rescorla, R. A. (1980b). *Pavlovian second-order conditioning*. Hillsdale, NJ: Erlbaum.
- Rescorla, R. A. (1981). Simultaneous associations. In P. Harzem & M. D. Zeiler (Eds.), *Predictability, correlation, and contiguity* (pp. 47-80). New York: Wiley.
- Rescorla, R. A. (1983). Pavlovian conditioning: Its not what you think it is. *American Psychologist*, 43, 151-160.
- Rescorla, R. A., & Wagner, A. R. (1972). A theory of Pavlovian conditioning: Variations in the effectiveness of reinforcement and nonreinforcement. In A. H. Black & W. F. Prokasy (Eds.), *Classical conditioning II: Current research and theory* (pp. 64-99). New York: Appleton-Century-Crofts.
- Rizley, R. C., & Rescorla, R. A. (1972). Associations in second-order conditioning and sensory preconditioning. *Journal of Comparative and Physiological Psychology*, 81, 1-11.
- Shurtleff, D., & Ayres, J. J. B. (1981). One-trial backward excitatory fear conditioning in rats: Acquisition, retention, extinction, and spontaneous recovery. *Animal Learning and Behavior*, 9, 65-74.
- Siegal, S., & Domjan, M. (1971). Backward conditioning as an inhibitory procedure. *Learning and Motivation*, 2, 1-11.
- Siegal, S., & Domjan, M. (1974). The inhibitory effect of backward conditioning as a

- function of the number of backward pairings. *Bulletin of the Psychonomic Society*, 4, 122-124.
- Spear, N. E. (1978). *The processing of memories: Forgetting and retention*. Hillsdale, NJ: Erlbaum.
- Speers, M. A., Gillan, D. J., & Rescorla, R. A. (1980). Within-compound associations in a variety of compound conditioning procedures. *Learning and Motivation*, 11, 135-149.
- Spetch, M. L., Wilkie, D. M., & Pinel, J. P. J. (1981). Backward conditioning: A reevaluation of the empirical evidence. *Psychological Bulletin*, 89, 163-175.
- Timberlake, W. (1986). Unpredicted food produces a mode of behavior that affects rats' subsequent reactions to a conditioned stimulus: A behavior systems approach to "context-blocking". *Animal Learning and Behavior*, 14, 276-286.
- Tolman, E. C., & Honzik, C. H. (1930). Introduction and removal of reward, and maze performance in rats. *University of California Publications in Psychology*, 4, 257-275.
- Wagner, A. R. (1976). Priming in STM: An information processing model for self-generated or retrieval-generated depression in performance. In T. J. Tighe & R. N. Leaton (Eds.), *Habituation: Perspectives from child development, animal behavior, and neuropsychology* (pp. 95-128). Hillsdale, NJ: Erlbaum.

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